



Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir
M.V. College
Of Science & Commerce

Department of Information Technology (Bsc.IT Semester III)
DATA STRUCTURE PROJECT

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Date of Assignment : 02/09/2026	Date/Time of Submission : 09/09/2026

DATA STRUCTURE PROJECT:

CODE:

Project DSA
Topic: Library Management System

```
books = []
```

```
# Array - Insert  
books.append("Science")  
books.append("Math")  
books.append("English")  
books.append("Hindi")
```

```
print("Books:", books)
```

```
# Array - Select  
print("Selected Book:", books[1])
```

```
# Array - Delete  
books.remove("Hindi")  
print("After Delete:", books)
```

```
# Linear Search
def linearsearch(array, n, x):
    for i in range(0, n):
        if array[i] == x:
            return i
    return -1

x = "English"
n = len(books)
result = linearsearch(books, n, x)

if result == -1:
    print("Book not Found")
else:
    print("Book found at index:", result)
```

```
# Queue - Enqueue
queue = []
queue.append("History")
queue.append("Civics")
queue.append("Geography")

print("Book Queue:", queue)
```

```
# Queue - Dequeue
print("Dequeue:", queue.pop(0))
print("Remaining Queue:", queue)
```

```
# Insertion Sort
def insertionSort(array):

    for step in range(1, len(array)):
        key = array[step]
        j = step - 1

        while j >= 0 and key < array[j]:
            array[j + 1] = array[j]
            j = j - 1

        array[j + 1] = key
```

```
insertionSort(books)
```

```
print("Sorted Books in Ascending Order:")
```

```
print(books)
```

```
print("Rahul_51")
```

OUTPUT:

```
===== RESTART: C:/Users/IT-!  
Books: ['Science', 'Math', 'English', 'Hindi']  
Selected Book: Math  
After Delete: ['Science', 'Math', 'English']  
Book found at index: 2  
Book Queue: ['History', 'Civics', 'Geography']  
Dequeue: History  
Remaining Queue: ['Civics', 'Geography']  
Sorted Books in Ascending Order:  
['English', 'Math', 'Science']  
Rahul_51
```